

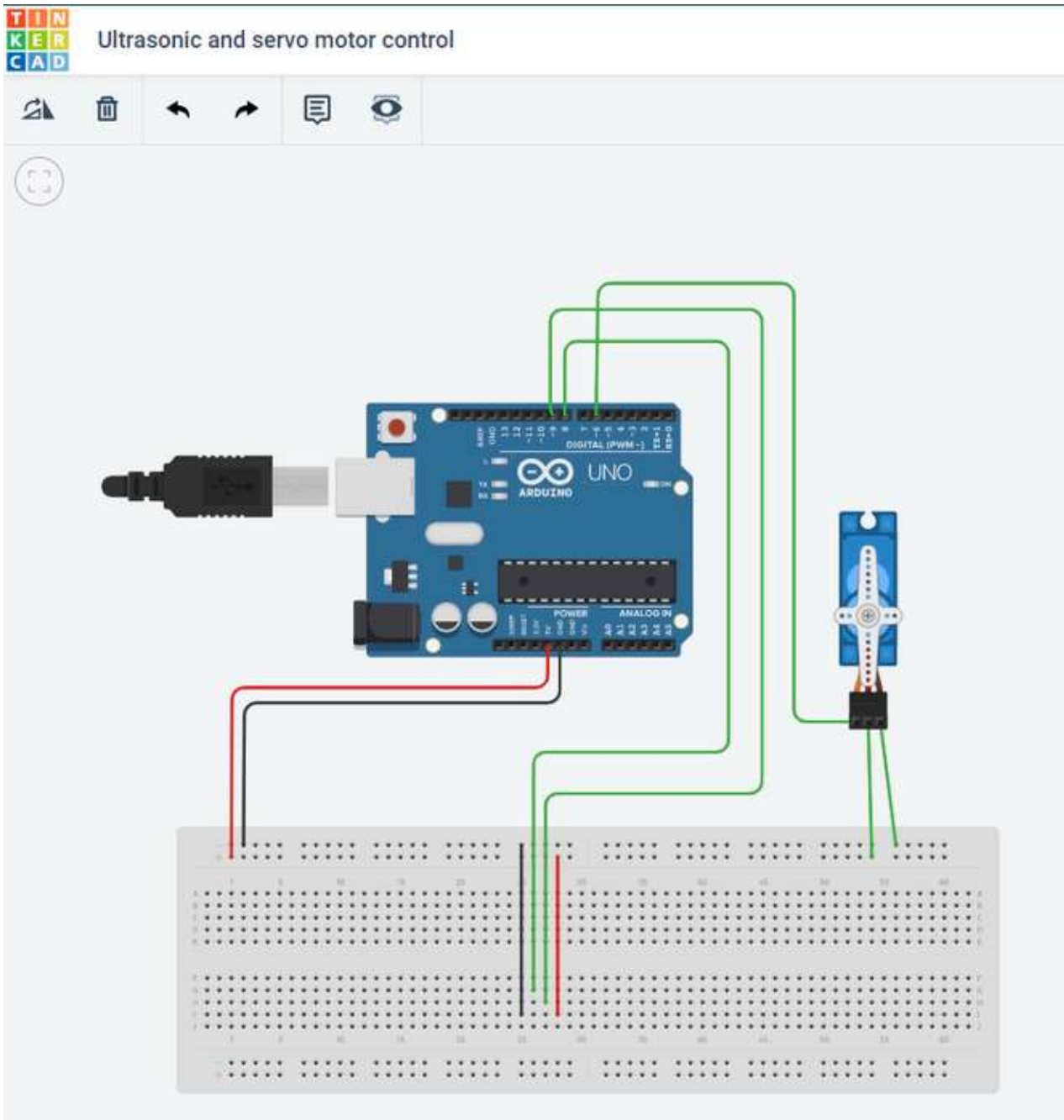
**AUTODESK**  
Instructables

# Controlling a Servo With an Ultrasonic Sensor Using Arduino

By [LevinCharles](#) in [CircuitsArduino](#)



## Introduction: Controlling a Servo With an Ultrasonic Sensor Using Arduino

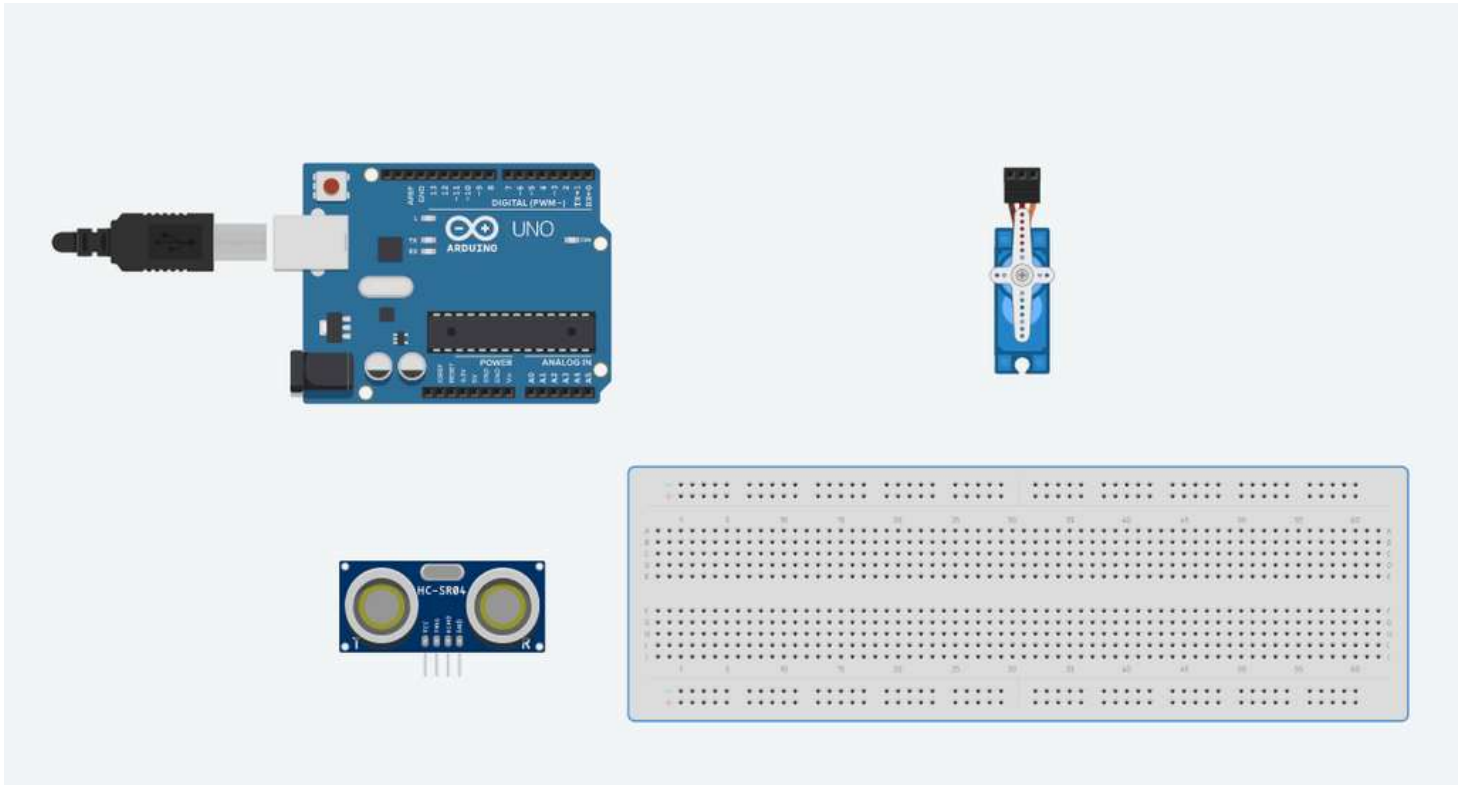


[Tinkercad Projects »](#)

We are a group of UQD10801 (Robocon1) students from Universiti Tun Hussein Onn Malaysia (UTHM) that will demonstrate how to control a servo motor using an ultrasonic sensor and an arduino as part of our assignment.

An ultrasonic sensor is a device that can measure the distance to an object by the use of sound waves while a servo motor is built for precision control of the angular or linear position, velocity, and acceleration.

## Step 1: Step 1: Components Needed



- 1) Arduino Uno x 1
- 2) Ultrasonic Sensor x 1
- 3) Servo Motor x 1
- 4) Breadboard

## Step 2: Step 2: Platform Used



The platform used to build the circuit and test the simulation was Tinkercad.

## Step 3: Step 3: the Working Principle

**Ultrasonic Distance Sensor**  
 Name

**Text**  

```

1  #include <Servo.h>
2  #define trigPin 9
3  #define echoPin 8
4  Servo servo;
5  int sound = 250;
6  void setup() {
7    Serial.begin (9600);
8    pinMode(trigPin, OUTPUT);
9    pinMode(echoPin, INPUT);
10   servo.attach(6);
11 }
12 void loop() {
13   long duration, distance;
14   digitalWrite(trigPin, LOW);
15   delayMicroseconds(2);
16   digitalWrite(trigPin, HIGH);
17   delayMicroseconds(10);
18   digitalWrite(trigPin, LOW);
19   duration = pulseIn(trigPin, HIGH);
20   distance = (duration * 0.034) / 2;
21   if (distance < 80) {
22     Serial.print(distance);
23     Serial.println(" cm");
24     servo.write(90);
25   }
26   else if (distance < 100) {
27     Serial.print(distance);
28     Serial.println(" cm");
29     servo.write(180);
30   }

```

**Serial Monitor**  

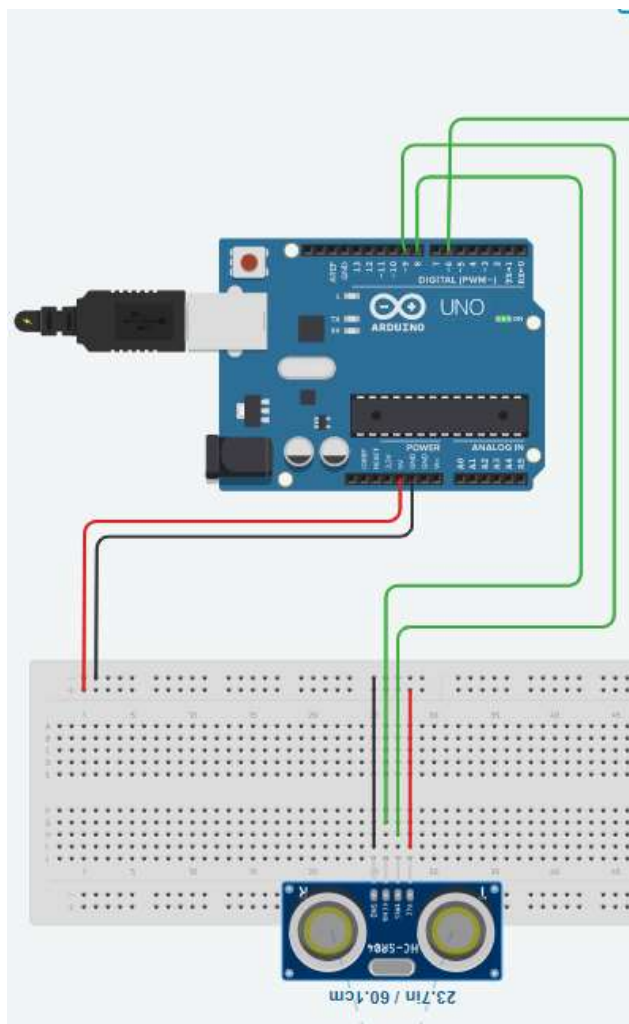
```

59 cm
59 cm
59 cm
59 cm
59 cm
59 cm
59 cm

```

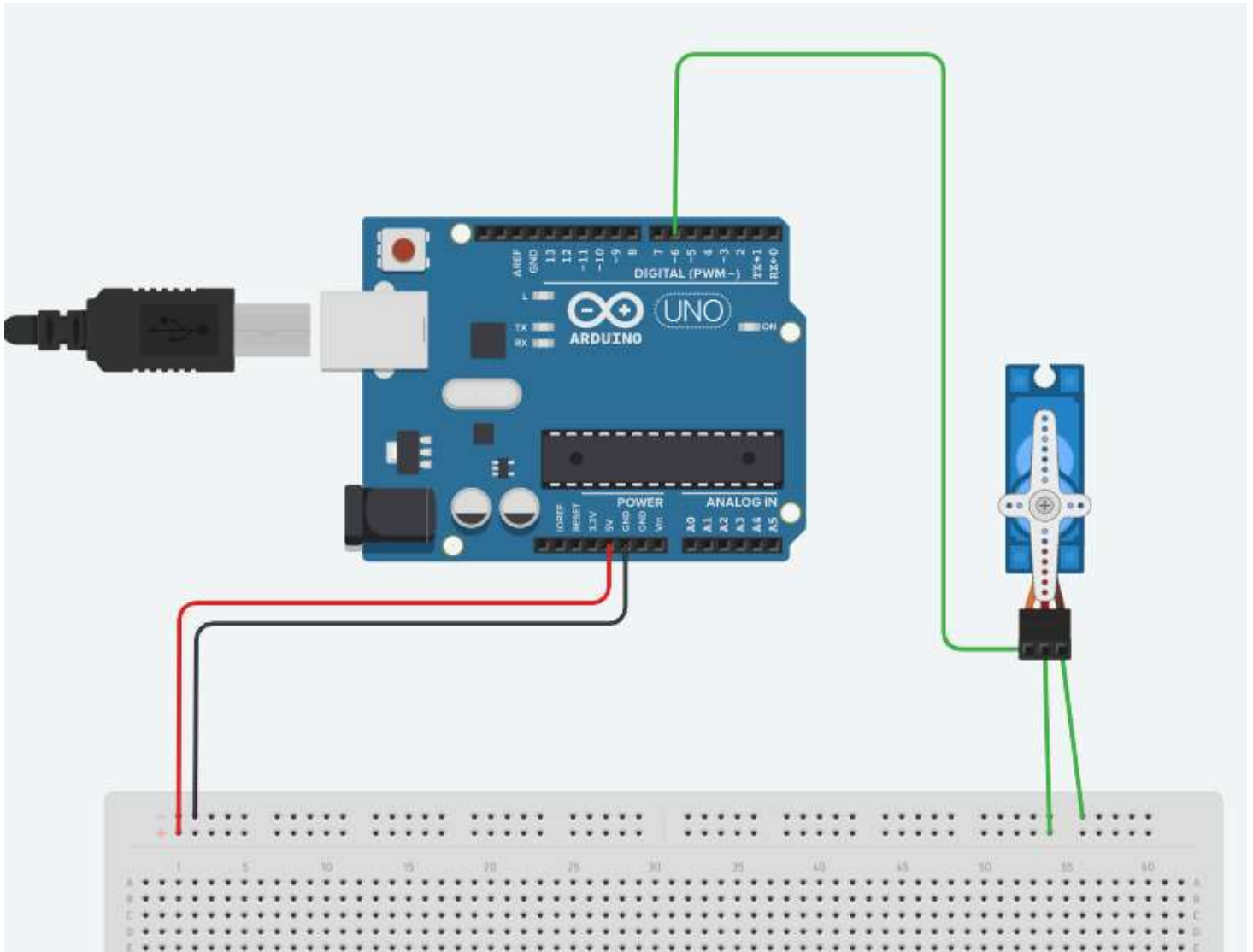
The servo motor will rotate to a certain angle depending on the distance detected by the ultrasonic sensor. The distance detected will also be displayed on the serial monitor.

## Step 4: Step 4: Connection (Ultrasonic Sensor)



- 1) Connect the 5V pin of the Arduino board to the bottom row of the breadboard
- 2) Connect the GND pin to the top row of the breadboard.
- 3) Connect the VCC pin of the sensor to the positive terminal of the breadboard.
- 4) Connect the GND pin to the negative terminal of the breadboard.
- 5) Connect the TRIG to pin 9 of the Arduino.
- 6) Connect the ECHO pin to pin 8 of the Arduino

## Step 5: Step 5: Connection (Servo Motor)



- 1) Connect the positive terminal/power to the positive terminal of breadboard.
- 2) Connect the ground terminal of servo motor to the negative terminal of the breadboard.
- 3) Connect the data wire/ signal to pin 6 of the Arduino board.

## Step 6: Step 6: Coding

```
#include <Servo.h>

#define trigPin 9

#define echoPin 8

Servo servo;

int sound = 250;

void setup() {
  Serial.begin (9600);

  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  servo.attach(6);
}

void loop() {
  long duration, distance;
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);
  duration = pulseIn(echoPin, HIGH);
  distance = (duration/2) / 29.1; if (distance < 80) {
    Serial.print(distance);
    Serial.println(" cm");
    servo.write(90);
  }
  else if (distance<180) {
    Serial.print(distance);
    Serial.println(" cm");
```

```
servo.write(180); }  
  
else {  
  Serial.println("The distance is more than 180cm");  
}  
  
delay(500);  
}
```

## Step 7: Tutorial Video

This video is made for better understanding of the overall process.